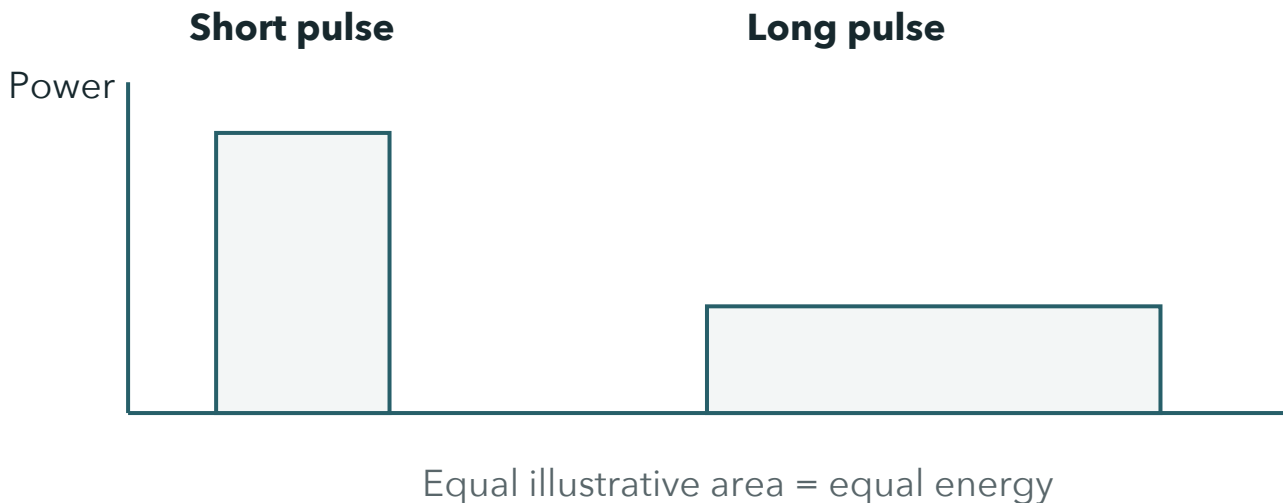


Laser dose and tissue interaction

Part 1 | Dose needs an area and a timescale

Equal joules do not guarantee the same tissue temperature history.



Equal pulse energy delivered over different durations has different peak power; thermal spread also depends on duration.

Two useful quantities

Irradiance is power per area; radiant exposure is energy per area. For a uniform constant-power exposure, radiant exposure = irradiance x duration. A shorter pulse with equal area and energy has higher average power during that pulse, but biological response also depends on absorption and heat transport.

A worked dose comparison

Keep spot area, duration and energy separate throughout the arithmetic.

Original example; uniform 0.01 square-centimetre spot

Exposure	Irradiance	Radiant exposure
0.2 W for 0.1 s	20 W/cm ²	2 J/cm ²
0.1 W for 0.2 s	10 W/cm ²	2 J/cm ²

What matches

The radiant exposures match but their power-time histories differ. Whether the lesions match cannot be decided from energy alone: the question must specify the thermal regime, target and absorption. Also distinguish corneal-plane machine settings from the actual retinal spot size and energy.

Thermal confinement has a characteristic time

Heat diffuses farther when the target has more time to exchange energy with its surroundings.

Selective-heating model

- Choose a preferentially absorbing target
- ↓
- Deliver energy within the relevant relaxation time
- ↓
- Limit heat spread into adjacent tissue
- ↓
- Assess target and neighbouring tissue responses

Experimental interpretation

Compare equal absorbed energy and target size while varying pulse duration. A shorter exposure can confine heating better than a prolonged exposure in the appropriate regime. This is a physical principle, not a universal recommendation to shorten every ophthalmic laser pulse.

Photochemical, thermal and disruptive effects differ

The operative mechanism depends on wavelength, absorption and temporal delivery.

Interaction

Feature	Main event	Typical distinction
Thermal	Heating and protein effects	Heat transport matters
Photochemical	Light-driven chemistry	Sensitiser/oxygen can matter
Photodisruptive	Optical breakdown; mechanical effects	High local intensity matters

A mechanism question

Do not identify a tissue effect solely from the word laser. Ask which material absorbs, whether a sensitiser is present, how concentrated the delivery is and what duration is used. The same broad technology can produce different dominant interactions under different conditions.

Safety follows the beam path

An invisible beam can still reach a susceptible ocular target.

Assessment

Feature	Question	Reason
Wavelength	Which tissue absorbs it?	Target differs by spectrum
Delivery	Where is the beam focused?	Local exposure differs
Protection	Is eyewear suited to this wavelength?	Generic tint is insufficient

Applied interpretation

Protective equipment is selected for the relevant wavelength and exposure, not merely because the lens appears dark. When interpreting a laboratory injury, reconstruct the optical path and actual beam geometry before using the instrument's nominal total power as the complete explanation.